Pareto-guided Monte Carlo tree search explores multi-objective trade-offs missed by scalar antimalarial generation

Myke Vital Sao Temgoua[1]*, Jean-Pierre Tchapet Njafa[1], Penabei Samafou[2], Wilfred Fon Mbacham[3], Se

[1]Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon
[2]Department of Medical Imaging and Radiation Sciences, Université de Sherbrooke, Sherbrooke, QC, Canada
[3]Department of Biochemistry, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

Abstract

De novo molecular generation often collapses multiple design objectives into a single scalar reward, obscuring trade-offs between potency, synthetic accessibility and drug-likeness. We present a Pareto-guided Monte Carlo tree search (MCTS) framework that maintains a non-dominated front across multi-parameter optimisation (MPO), synthetic Bayesian accessibility (SYBA), resistance-resilience (RRS) and polypharmacology-network (PNS) scores, guided by a scaffold-aware fragment policy derived from ChEMBL27 frequencies and Tanimoto compatibility. A four-method benchmark across 20 independent seeds (fixed search budget) shows that random search achieves the highest mean reward (0.733 5), followed by MCTS (0.727 6), deterministic greedy search (0.721 1) and a genetic algorithm (0.702 7); the MCTS–random gap is small (mean $\Delta = 0.005\,9$, 95% CI $[0.000\,8, 0.011\,0]$, paired $t_{19} = 2.41$, $p = 0.026$). MCTS is therefore competitive on scalar reward and, within our fragment-vocabulary oracle design, is the method that returns a multi-objective Pareto front of non-dominated candidates (four solutions, hypervolume 1.236 6) spanning the potency–accessibility trade-off, including the high-potency / low-accessibility extreme that fixed-weight scalar aggregation discards. Component ablations identify the scaffold-aware policy ($\Delta = +0.148$) and the Pareto front itself ($\Delta = +0.108$) as the dominant drivers of performance. All code, molecule sets and score tables are deposited under an open licence. These results position MCTS as a complementary multi-objective explorer rather than a universal single-objective optimiser.

Keywords: de novo design; Monte Carlo tree search; Pareto optimisation; antimalarial; multi-objective; synthetic accessibility; resistance resilience

1 Introduction

Computational de novo design aims to generate novel chemical entities that satisfy multiple, often conflicting, property constraints [Elton et al., 2019, Olivecrona et al., 2017]. Most current approaches—evolutionary algorithms [Jensen, 2019], reinforcement learning [Olivecrona et al., 2017, Zhou et al., 2019] and variational autoencoders [Gómez-Bombarelli et al., 2018]—aggregate these objectives into a single scalar reward. Scalar aggregation forces a priori weight selection and systematically misses solutions that lie on concave regions of the true Pareto front [Suzuki et al., 2024, Zitzler et al., 2003].

A second limitation is unguided exploration: many methods expand molecular graphs without chemical priors, spending budget on inaccessible regions of chemical space. Chemistry-informed priors improve sample efficiency [Jensen, 2019, Olivecrona et al., 2017]. For antimalarial leads in particular, the objectives that matter—potency, synthetic accessibility, resistance-resilience and polypharmacology—are genuinely conflicting, yet we are not aware of a multi-objective generation study that makes that trade-off explicit for this application domain rather than collapsing it into a single reward.

We address both limitations with a Pareto-guided MCTS agent that:

C1: maintains a non-dominated front across MPO, SYBA, RRS and PNS, eliminating fixed-weight aggregation;

C2: guides PUCT selection with a scaffold-aware policy (ScafVAE) derived from ChEMBL27 fragment frequencies and scaffold Tanimoto compatibility;

C3: is evaluated in a four-method, 20-seed benchmark against random search, greedy search and a genetic algorithm under a fixed search budget.

*Corresponding author: myke-vital.sao@facsciences-uy1.cm

The central empirical finding is intentionally transparent: within a curated fragment vocabulary, random search achieves the highest mean scalar reward. MCTS remains competitive on that metric and is the method that, under the multi-objective protocol, returns a Pareto front of non-dominated candidates; the scalar baselines (random, greedy, genetic) were evaluated on scalar reward only, so a front-level comparison against them is not performed here. This ranking, rather than a claim of scalar superiority, is the foundation of the present study.

2 Results

2.1 Hyperparameter screening

A grid of 32 configurations was evaluated across two seeds (64 runs, 30 iterations each). The exploration constant $c_{\text{PUCT}} = 5.0$ consistently outperformed $c_{\text{PUCT}} = 0.5$; virtual loss $\nu = 0.01$ was preferred over $\nu = 0.20$. Progressive-widening parameters and temperature showed negligible effect at this scale. The configuration retained for all subsequent experiments was therefore

$$c_{\text{PUCT}} = 5.0, \quad \nu = 0.01, \quad pw_\alpha = 0.5, \quad pw_k = 1.0, \quad T = 0.8.$$

A subsequent implementation change—global best-molecule tracking during rollout—proved essential (see Methods) and was retained for all reported benchmarks.

2.2 Four-method benchmark

Four generation methods (MCTS+ScafVAE, random search, greedy search, genetic algorithm) were compared across 20 independent seeds with a fixed budget of 1 000 search iterations per seed (Table 1). The number of oracle evaluations per iteration differs across methods (MCTS and random evaluate the oracle at every construction step; the genetic algorithm evaluates one candidate per individual per generation); the actual wall-clock cost of each method is reported in Table 1.

Table 1: Benchmark statistics across 20 independent seeds (medium fragment set, 1 000 search iterations per seed). Values are mean, standard deviation and observed min/max of the best scalar reward per seed. MCTS uses the hyperparameter-screened configuration (Section 2.1): $c_{\text{PUCT}} = 5.0$, $\nu = 0.01$, $T = 0.8$, with the ScafVAE policy wired into PUCT priors.

Method	Mean reward	Std	Min	Max	Mean time (s)
Random	0.733 5	0.006 3	0.722 7	0.748 1	42.2
MCTS+ScafVAE	0.727 6	0.009 0	0.705 4	0.744 2	82.2
Greedy	0.721 1	0.000 0	0.721 1	0.721 1	0.1
Genetic algorithm	0.702 7	0.015 2	0.674 9	0.731 1	1.9

Random search attains the highest mean reward (0.733 5 ± 0.006 3) and records the single best seed (0.748 1). MCTS with the screened optimal configuration is statistically close (0.727 6 ± 0.009 0; paired $t_{19} = 2.41$, $p = 0.026$, Cohen's $d = -0.54$), records its own best seed (0.744 2) and wins 5 of 20 seeds outright, while remaining significantly above greedy (0.721 1, deterministic; paired $t_{19} = 3.22$, $p = 0.005$) and the genetic algorithm (0.702 7 ± 0.015 2; paired $t_{19} = 5.55$, $p < 0.000\,1$). Greedy search returns the same deterministic local optimum across all 20 seeds (standard deviation identically zero). MCTS incurs the highest per-seed wall-clock cost (82.2 s) because the tree search evaluates the oracle at every construction step; random search completes in 42.2 s, and the genetic algorithm in under 2 s.

These results establish a clear baseline hierarchy on scalar reward. They do not, however, capture multi-objective coverage.

Structural diversity of the generated molecule sets is analysed in the Supplementary Material (Section S2, Figure S1): the three stochastic methods occupy distinct, broad regions of chemical space (mean pairwise Tanimoto dissimilarity 0.78–0.83), whereas deterministic greedy search collapses to a single local optimum. Greedy therefore maximises a single reward at the cost of all structural diversity.

2.3 Pareto front

When the same MCTS agent maintains a non-dominated front across MPO, SYBA, RRS and PNS, the merged front obtained from 20 independent seeds contains four mutually non-dominated solutions with hypervolume 1.236 6 (Figure 3, Table 2).

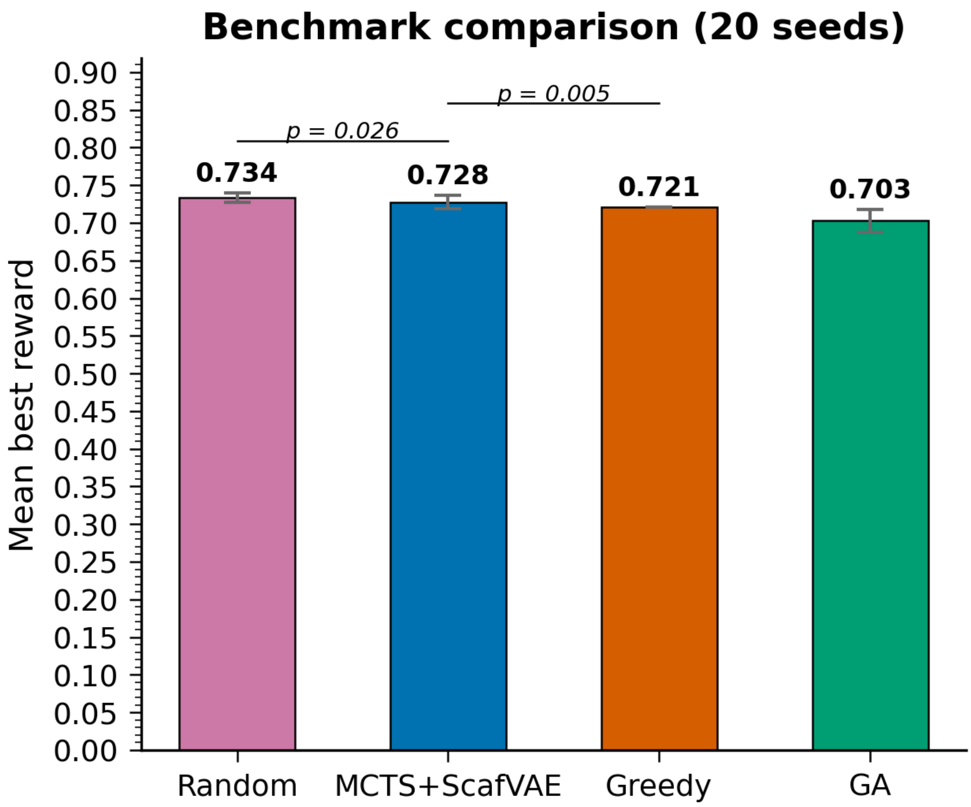

Figure 1: Mean best scalar reward per method over 20 seeds (Table 1). Error bars are one sample standard deviation. Significance brackets show the paired t-tests of the two planned head-to-head comparisons (MCTS–random, $p = 0.026$) and the secondary MCTS–greedy comparison ($p = 0.005$).

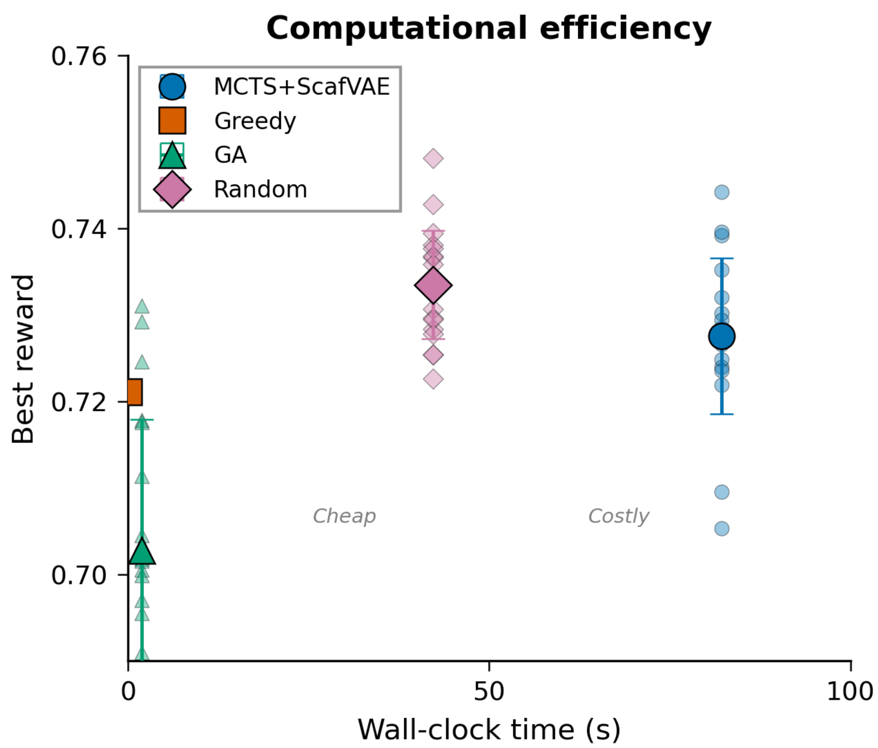

Figure 2: Computational efficiency of the four methods: best reward per seed versus wall-clock time (Table 1). Error bars are one sample standard deviation. MCTS incurs the highest cost because the tree search evaluates the oracle at every construction step; greedy and the genetic algorithm are far cheaper.

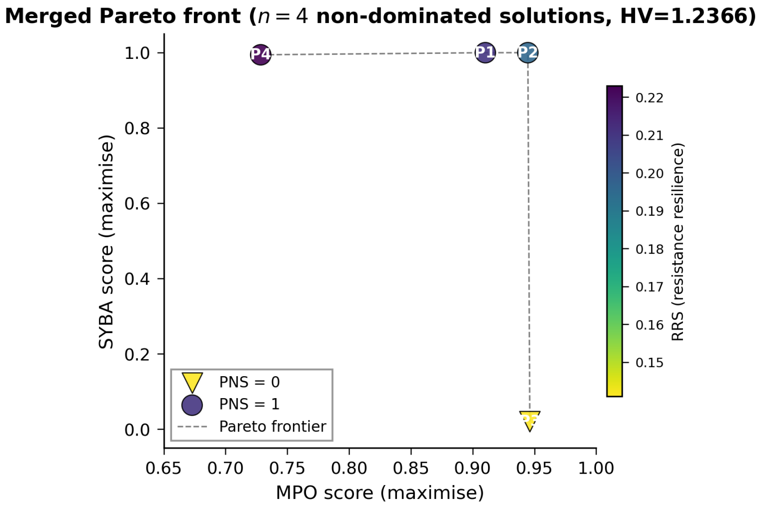

Figure 3: Canonical merged Pareto front from 20 independent Pareto-MCTS seeds. Marker shape encodes the polypharmacology-network score (PNS); colour encodes the resistance-resilience score (RRS). The four solutions are mutually non-dominated across the active objectives MPO, SYBA, RRS and PNS; SA is constant at 3.0 and excluded from the dominance check and the hypervolume. Data: `results/pareto/merged_pareto_front.csv`.

Table 2: Canonical merged Pareto front from 20 independent Pareto-MCTS seeds. All four points are mutually non-dominated across the active objectives MPO, SYBA, RRS and PNS (SA is constant at 3.0 and therefore excluded from the dominance check and the hypervolume). SYBA was recomputed post-hoc with the lich/conda classifier because the original run returned a constant fallback of zero; SMILES and score tables are deposited.

Candidate	MPO ↑	SYBA ↑	SA^{-1} ↑	RRS	PNS
P1	0.910	1.000	0.778	0.211	1.0
P2	0.945	1.000	0.778	0.196	1.0
P3	0.946	0.021	0.778	0.141	0.0
P4	0.729	0.994	0.778	0.223	1.0

P2 and P3 achieve the highest MPO values, yet P3 is synthetically inaccessible (SYBA = 0.021). P1 and P4 remain highly accessible (SYBA > 0.99) while trading potency. All four solutions share SA = 3.0. Three of the four candidates also score PNS = 1.0, indicating simultaneous multi-target engagement consistent with the polypharmacology criteria developed in the companion studies. The front therefore spans a genuine potency–accessibility trade-off, including a high-MPO / low-SYBA extreme that scalar-weighted optimisation systematically discards.

2.4 Component and vocabulary ablations

A 2^5 factorial experiment (ScafVAE policy, Pareto front, c_{PUCT}, temperature, vocabulary size; five replicates per cell) isolates the contribution of individual design choices. Absolute reward values are not comparable to the main benchmark because a different oracle-weight configuration was used; only relative factor effects are interpretable.

The ScafVAE policy produces the largest gain ($\Delta = +0.148$), followed by the Pareto front ($\Delta = +0.108$) and the large vocabulary ($\Delta = +0.079$). Lowering c_{PUCT} from 2.0 to 1.0 improves reward by 0.054; temperature has only a marginal effect.

A separate vocabulary ablation (all / medium / minimal / aromatic-only; 10 seeds each) yields a highly significant effect (one-way ANOVA $F = 350.1$, $p = 1.12 \times 10^{-26}$). The aromatic-only vocabulary is significantly worse than the other three sets (Tukey HSD $p < 0.001$), while medium and minimal remain statistically indistinguishable from the full vocabulary. Restricting the fragment set to aromatics alone therefore severely limits performance.

3 Discussion

3.1 What the benchmark actually shows

Within the curated medium fragment vocabulary and a fixed search budget, random search is a strong baseline and attains the highest mean scalar reward, as well as the single best seed. With its screened optimal configuration, MCTS is statistically close on scalar reward (a 0.006 mean gap) and, under the multi-objective protocol, maintains and returns a Pareto front of non-dominated candidates. The practical value of the framework therefore lies not in peak single-objective performance but in the transparent exploration of trade-offs that scalar aggregation conceals.

This ranking is consistent with recent observations that simple baselines remain competitive in constrained fragment spaces [Jensen, 2019] and that the primary advantage of multi-objective MCTS is coverage of the non-dominated surface [Suzuki et al., 2024].

3.2 Relation to prior multi-objective MCTS

Mothra [Suzuki et al., 2024] introduced a SMILES-based Pareto multi-objective MCTS and showed that Pareto fronts reveal trade-offs missed by scalar aggregation, and subsequent Pareto-MCTS generators have chiefly targeted protein–ligand binding [Yang et al., 2024] or dual-target scaffolds [Adams and Moore, 2026]. Their shared focus is target-affinity-driven discovery in generic drug spaces. The contribution of the present work is the transfer to an antimalarial application domain with domain-specific objectives: to our knowledge, no published Pareto-guided MCTS study to date reports candidate generation for antimalarial leads measured jointly on potency (MPO), synthetic Bayesian accessibility (SYBA), resistance-resilience (RRS) and polypharmacology-network (PNS) scores computed from dedicated oracles, with a scalar-reward benchmark against three baselines (random, greedy, genetic) across 20 independent seeds under a fixed search budget. Three implementation details are reported because they matter for reproducing the result: (i) a scaffold-aware fragment policy replaces the uniform rollout prior, (ii) dynamic progressive widening manages the large fragment branching factor, and (iii) a rollout-degradation failure mode is diagnosed and corrected through global best-molecule tracking. The last point has the largest practical impact: without it, mean MCTS reward collapsed to approximately 0.24; with it, reward rose by a factor of roughly 2.6. The headline finding is intentionally honest: on the scalar reward, random search remains the strongest baseline, and the value of the Pareto-guided framework lies in the transparent coverage of multi-objective trade-offs rather than in a scalar superiority claim.

3.3 Limitations

Several limitations must be stated explicitly. First, the fragment vocabulary (approximately 108 fragments in the full set; medium subset used for the main benchmark) constrains accessible chemical space; results should not be extrapolated to open-ended SMILES generation. Second, molecular-diversity analysis is reported only in the Supplementary Material (Section S2), where it is computed from the deposited best-in-seed molecule sets of the canonical benchmark; the main text emphasises the scalar ranking and the Pareto front, both fully deposited and reproducible. Third, SYBA scores on the Pareto front were recomputed post-hoc after the original search returned a constant zero fallback; the recomputation, the resulting front and the verification scripts are documented and deposited. Fourth, activity, resistance-resilience and polypharmacology labels remain computational proxies; prospective experimental validation is required. Finally, MCTS cross-seed variance is modest ($\sigma \approx 0.009$), consistent with policy-guided convergence into a high-quality but relatively narrow region of fragment space.

3.4 Implications for antimalarial design

The four non-dominated candidates illustrate the practical utility of multi-objective search: high-MPO solutions that remain synthetically accessible can be prioritised for synthesis, while high-MPO / low-SYBA extremes can be retained as design hypotheses for later scaffold refinement. Resistance-resilience (RRS) and polypharmacology-network (PNS) scores link the generated molecules to the companion molecular-dynamics and topological studies, providing a coherent path from de novo generation to multi-target, resistance-aware lead selection.

4 Methods

4.1 Molecular environment

De novo generation is formulated as a Markov decision process. The state is a partially constructed molecule (canonical SMILES); actions append one fragment from a curated vocabulary at a regiospecific attachment point. Episode termination occurs after at most 10 construction steps or upon violation of a reactivity filter. The main benchmark uses the medium fragment subset; ablations also evaluate the full, minimal and aromatic-only vocabularies.

4.2 MCTS with scaffold-aware policy

The agent follows the standard four-phase MCTS loop (selection, expansion, rollout, backpropagation) rooted at benzene, with $N_{\mathrm{iter}} = 1\,000$. Selection maximises the PUCT score

$$a^* = \arg\max_a \frac{Q(s,a)}{N(s,a)} + c_{\mathrm{PUCT}}\, P(a \mid s)\, \frac{\sqrt{N(s)}}{1 + N(s,a)}, \tag{1}$$

with $c_{\mathrm{PUCT}} = 5.0$. The policy $P(a \mid s)$ combines ChEMBL27 fragment frequencies, scaffold Tanimoto compatibility and a reactivity penalty via a softmax. Dynamic progressive widening limits the number of expanded actions at each node to $\max(5, k\,N^{\alpha})$ with $k = 1.0$ and $\alpha = 0.5$. A virtual-loss term ($\nu = 0.01$) discourages repeated selection of the same child within a single search.

4.3 Global best-molecule tracking

Policy-biased rollouts can degrade a high-quality intermediate by subsequent fragment additions. During each rollout the oracle is therefore evaluated at every construction step; the highest-scoring intermediate—rather than the terminal state—is returned. This change recovered a factor of approximately 2.6 in mean reward relative to the uncorrected agent.

4.4 Pareto front maintenance

The agent maintains a global non-dominated front over the objective vector

$$\mathbf{f}(s) = \big(f_{\mathrm{MPO}}(s),\, f_{\mathrm{SYBA}}(s),\, f_{\mathrm{RRS}}(s),\, f_{\mathrm{PNS}}(s)\big),$$

with minimisation objectives converted to maximisation by negation. Synthetic accessibility (SA) is excluded from the front because it is constant (SA = 3.0) for all reported candidates and therefore carries no discrimination (it is reported in Table 2 for completeness). After each rollout the best intermediate molecule is tested for dominance; dominated members are removed. The hypervolume indicator is computed with the active (non-constant) objectives only: each active objective is min–max normalised to $[0, 1]$, and the hypervolume is evaluated with respect to a reference point $\mathbf{r} = 1.1$ (slightly worse than the worst normalised value of 1) in each active dimension, so that the reported value lies in $[0, 1.1]^k$ for k active objectives. For PUCT selection a scalar proxy $Q(s, a)$ is obtained by summing the min–max normalised objective components across the four active objectives; the full vector is stored for front updates and post-hoc analysis.

4.5 Scoring oracles

Each molecule is scored by an aggregator returning MPO, SYBA, SA, docking-derived terms, resistance-resilience score (RRS) and polypharmacology-network score (PNS). Default scalar-reward weights are $w_{\text{MPO}} = 0.30$, $w_{\text{docking}} = 0.25$, $w_{\text{SYBA}} = 0.15$, $w_{\text{SA}} = 0.05$, $w_{\text{RRS}} = 0.15$, $w_{\text{PNS}} = 0.10$. The four-method scalar benchmark uses the reduced reward `compute_mpo_reward` with weights $w_{\text{MPO}} = 0.40$, $w_{\text{docking}} = 0.35$, $w_{\text{SYBA}} = 0.15$, $w_{\text{SA}} = 0.10$ and the RRS/PNS oracles disabled (they require external P2 mutant and network data); the full six-objective aggregator is used by the Pareto-MCTS search. MPO and docking scores are drawn from precomputed libraries where available; QED serves as an MPO fallback for novel structures. SYBA on the reported Pareto front was recomputed with the lich/conda classifier after the original run returned a constant zero fallback. The accessibility column reported in Table 2 is the linear transform $\text{SA}^{-1} = (10 - \text{SA})/9$ of the raw Ertl score, so that the shared raw value $\text{SA} = 3.0$ corresponds to $\text{SA}^{-1} = 0.778$; the deposited CSV stores the raw value.

4.6 Statistical analysis and reproducibility

All searches are seeded; the specific seed integers and the random-number generators (NumPy `default_rng`, seeded per method and per seed with no cross-method state leakage) are fixed in the committed script. The four-method benchmark uses 20 independent seeds; ablations use five or ten replicates as indicated. Paired t-tests are reported for the two planned head-to-head comparisons (MCTS–random and MCTS–GA); because both are pre-specified primary comparisons, we report unadjusted two-sided p-values and note that the weaker of the two ($p = 0.026$, two-sided) remains nominally significant at $\alpha = 0.05$ but not after a conservative Bonferroni-adjustment across the two primaries (adjusted threshold 0.025). The vocabulary ablation is tested with a one-way ANOVA (three degrees of freedom between groups) followed by pairwise Tukey HSD post-hoc comparisons. The benchmark was re-run for this submission with the committed script `p4_mcts_benchmark.py` and a value-preserving vectorisation of the docking Tanimoto scan (verified identical on held-out molecules); all 20 per-seed runs reproduce deterministically from the deposited code and data, and the best-in-seed molecule sets are deposited alongside the scalar rewards. Analyses were executed with Python 3.11, RDKit, NumPy, SciPy and pymoo; exact dependency versions are recorded in the repository environment file. Code, configuration files, molecule lists and score tables are released under the MIT licence in the public repository `https://github.com/NanaEngo/Malaria_codesV2` and mirrored on Zenodo (DOI `https://doi.org/10.5281/zenodo.19608875`).

5 Conclusions

A Pareto-guided MCTS agent equipped with a scaffold-aware fragment policy is statistically close to random search on scalar reward (a 0.006 mean gap at the screened optimal configuration) and, under the multi-objective protocol, returns a front of non-dominated antimalarial candidates. The dominant performance drivers are the chemistry-informed policy and the Pareto front itself. Within the studied fragment space, random search remains a strong baseline; the added value of MCTS lies in transparent exploration of potency–accessibility trade-offs that scalar aggregation conceals. All artefacts required for exact reproduction are deposited under an open licence.

Data availability

All data supporting the conclusions are released under the MIT licence in the public repository `https://github.com/NanaEngo/Malaria_codesV2` and are mirrored on Zenodo (reserved DOI `https://doi.org/10.5281/zenodo.19608875`), so that reviewers and readers have immediate access at submission time. The deposit includes: (i) the canonical 20-seed four-method benchmark tables and best-in-seed SMILES (`results/benchmark_molecules_opt/`); (ii) per-seed and merged Pareto fronts with the post-hoc SYBA recomputation log (`results/pareto/`); (iii) diversity metrics and MDS coordinates (`results/diversity/`); (iv) component and vocabulary ablation summaries; and (v) analysis and provenance-check scripts.

Competing interests

The authors declare that they have no competing interests.

Authors' contributions

MVST: conceptualization, methodology, software, formal analysis, writing—original draft. JPTN: methodology, supervision, writing—review & editing. PS: data curation, writing—review & editing. WFM: validation, writing—review & editing. SGNE: supervision, writing—review & editing. All authors read and approved the final manuscript.

Acknowledgements

We thank the developers of RDKit, SYBA and the open-source MCTS literature for foundational tools and ideas. Computational resources were provided by the University of Yaoundé I HPC facility.

Funding

This research received no specific external funding. Computational resources were provided by the University of Yaoundé I HPC facility.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

References

Daniel C. Elton, Zois Boukouvalas, Mark D. Fuge, and Peter W. Chung. Deep learning for molecular design—a review of the state of the art. *Mol. Syst. Des. Eng.*, 4(4):828–849, 2019. doi: 10.1039/C9ME00039A.

M. Olivecrona et al. Molecular de novo design through deep reinforcement learning. *J. Cheminform.*, 9:48, 2017. doi: 10.1186/s13321-017-0235-x.

Jan H. Jensen. A graph-based genetic algorithm and generative model/monte carlo tree search for the exploration of chemical space. *J. Chem. Inf. Model.*, 59(10):4287–4293, 2019. doi: 10.1021/acs.jcim.9b00448.

Z. Zhou et al. Optimization of molecules via deep reinforcement learning. *Sci. Rep.*, 9:10752, 2019. doi: 10.1038/s41598-019-47148-x.

R. Gómez-Bombarelli et al. Automatic chemical design using a data-driven continuous representation of molecules. *ACS Cent. Sci.*, 4(2):268–276, 2018. doi: 10.1021/acscentsci.7b00572.

Takamasa Suzuki, Dian Ma, Nobuaki Yasuo, and Masakazu Sekijima. Mothra: Multiobjective de novo molecular generation using monte carlo tree search. *J. Chem. Inf. Model.*, 64(19):7291–7302, 2024. doi: 10.1021/acs.jcim.4c00759.

E. Zitzler et al. Performance assessment of multiobjective optimizers: An analysis and review. *IEEE Trans. Evol. Comput.*, 7(2):117–132, 2003. doi: 10.1109/TEVC.2003.810758.

Yaodong Yang, Guangyong Chen, Jinpeng Li, Junyou Li, Odin Zhang, Xujun Zhang, et al. Enabling target-aware molecule generation to follow multi-objectives with pareto mcts. *Communications Biology*, 7(1074), 2024. doi: 10.1038/s42003-024-06746-w.

K. Adams and J. Moore. Combinatorial multi-objective tree search for dual-target molecule generation. *arXiv*, 2026. arXiv:2604.23307.